IB Biology SL — 2025 Syllabus

THEME A - Unity & Diversity

Exam: Paper 1A (MCQ) · Paper 1B (Data) · Paper 2A (Short answer) · Paper 2B (Extended response)

WHAT'S IN THIS GUIDE

A1.1 Water — A1.2 Nucleic Acids — A2.2 Cell Structure — A3.1 Diversity of Organisms — A4.1 Evolution & Speciation — A4.2 Conservation of Biodiversity

This theme asks: what do all living things share? It covers the universal chemistry of life (water, nucleic acids, cells) and then explores the astonishing diversity built on that shared foundation.

High-frequency exam topics: water properties and their consequences, DNA structure, prokaryote vs eukaryote, magnification calculations, natural selection, and conservation strategies.

A1.1 — Water

The medium of life — properties and biological consequences

Water is the solvent of life. Its unique properties arise directly from its molecular structure — and these properties make life on Earth possible. Every property tested in this topic connects directly back to hydrogen bonding.

KEY DEFINITIONS

Polarity: water molecules have an unequal distribution of charge — the oxygen atom pulls electrons toward itself (it is more electronegative), creating a partial negative charge (δ−) on oxygen and partial positive charges (δ+) on each hydrogen.

Hydrogen bond: an attraction between the δ− oxygen of one water molecule and the δ+ hydrogen of a neighbouring water molecule. Each water molecule can form up to 4 hydrogen bonds. These bonds are individually weak but collectively very strong.

Cohesion: attraction between water molecules (due to H-bonding) — they 'stick together.' This creates surface tension.

Adhesion: attraction of water molecules to other polar surfaces (e.g. cellulose in xylem vessel walls) — water 'sticks' to surfaces.

Specific heat capacity: the energy needed to raise the temperature of 1g of a substance by 1°C. Water has a very HIGH specific heat capacity — it resists temperature change.

Latent heat of vaporisation: the energy needed to convert liquid water to vapour. Very HIGH for water — evaporation removes a lot of heat.

Hydrophilic: water-loving — polar or charged substances that dissolve in water (e.g. glucose, ions).

Hydrophobic: water-fearing — non-polar substances that do not dissolve in water (e.g. lipids, oils).

Property	Cause (H-bonding)	Biological consequence / importance
High specific heat capacity	Large amount of energy needed to break H-bonds and increase temperature — water absorbs much heat with little temperature rise	Aquatic environments: stable temperature — protects organisms. Blood and cytoplasm: resist rapid temperature changes. Oceans: regulate Earth's climate.
High latent heat of vaporisation	Breaking H-bonds requires large energy input — evaporation removes large amounts of heat	Sweating/panting: efficient cooling mechanism. Transpiration in plants: cools leaves. Evaporation from water bodies: moderates temperature.
Cohesion (surface tension)	H-bonds between water molecules pull them together	Insects walk on water surface. Cohesion-tension in xylem: pulls water column upward through plants (transpiration stream). Water forms droplets.
Adhesion	Water attracted to polar surfaces (e.g. xylem cell walls)	Assists capillary action — water travels up narrow xylem vessels against gravity.
Excellent solvent	Polar water molecules surround and dissolve polar/ionic substances (hydration shells)	Ions and polar molecules dissolved in cytoplasm for reactions. Transport of nutrients, gases, waste in blood and sap.
Density anomaly — ice less dense than liquid water	H-bonds in ice form a rigid lattice with molecules further apart than in liquid water	Ice floats on water → insulates aquatic environments → prevents complete freezing → aquatic life survives winter beneath ice.
Transparency	Water transmits visible light	Photosynthesis possible in aquatic environments — light reaches aquatic plants and algae.

EXAM TIP

When asked to 'explain a property of water', always state (1) the cause (hydrogen bonding / polarity), (2) what the property is, and (3) the biological consequence. One mark per point — be precise.

Common trap: 'Water is a good solvent because it is wet.' — This earns zero. Say: 'Water is a good solvent because its polar molecules surround and isolate ionic or polar solutes, forming hydration shells.'

Ice floats because it is LESS dense than liquid water — H-bonds in ice form an open lattice with molecules further apart than in liquid. This question appears often — know the reason, not just the fact.

A1.2 — Nucleic Acids

Nucleotides · DNA structure · RNA types · Antiparallel · Conservation of code

Nucleic acids (DNA and RNA) are the information molecules of life. DNA stores genetic information; RNA carries and uses it. Their structure is built from nucleotides, and understanding the structure is essential for understanding how DNA is replicated and how proteins are made (Themes D1.1 and D1.2).

KEY DEFINITIONS

Nucleotide: the monomer of nucleic acids. Each nucleotide has THREE components: (1) a phosphate group, (2) a pentose sugar (5-carbon), and (3) a nitrogenous base.

Pentose sugar: 5-carbon sugar. Deoxyribose in DNA (missing an oxygen at the 2' carbon). Ribose in RNA (has an —OH at the 2' carbon).

Nitrogenous base: the variable part of the nucleotide. Two categories: Purines (double ring — Adenine and Guanine) and Pyrimidines (single ring — Cytosine, Thymine [DNA only], Uracil [RNA only]).

Phosphodiester bond: covalent bond linking nucleotides together — between the phosphate of one nucleotide and the 3' carbon of the next sugar. Forms the sugar-phosphate backbone.

Complementary base pairing: A pairs with T (in DNA) or U (in RNA); C pairs with G. A–T: 2 hydrogen bonds. C–G: 3 hydrogen bonds.

Antiparallel: the two strands of DNA run in opposite directions — one 5'→3', the other 3'→5'. This is essential for replication and transcription.

5' end: the end of a polynucleotide strand with a free phosphate group. 3' end: the end with a free —OH group.

Feature	DNA	RNA
Sugar	Deoxyribose (missing —OH at 2' carbon)	Ribose (has —OH at 2' carbon)
Bases	Adenine (A), Thymine (T), Cytosine (C), Guanine (G)	Adenine (A), Uracil (U) [not T], Cytosine (C), Guanine (G)
Strands	Double-stranded — double helix	Single-stranded (usually)
Structure	Two antiparallel polynucleotide chains wound in a helix; H-bonds between complementary bases	Various shapes (mRNA = linear strand; tRNA = clover-leaf; rRNA = complex folded structure)
Location	Primarily nucleus (also mitochondria, chloroplasts)	Nucleus (transcription) and cytoplasm (translation)
Function	Long-term storage of genetic information	Carries and uses genetic information for protein synthesis
Types	One type (genomic DNA)	Three main types: mRNA (messenger), tRNA (transfer), rRNA (ribosomal)

RNA type	Full name	Structure	Function
mRNA	Messenger RNA	Linear single strand; sequence of codons (triplets)	Carries genetic code from nucleus to ribosome for translation
tRNA	Transfer RNA	Cloverleaf shape — 3D L-shape; anticodon loop at one end; amino acid attachment at 3' end	Brings specific amino acids to ribosome; anticodon pairs with mRNA codon
rRNA	Ribosomal RNA	Complex 3D structure; forms part of the ribosome (with proteins)	Structural and catalytic role in ribosome — peptidyl transferase activity

Conservation of the Genetic Code — Evidence for Common Ancestry

The genetic code (which codon codes for which amino acid) is virtually identical in ALL living organisms — from bacteria to humans.

This is extraordinary: there is no chemical reason why UUU must code for phenylalanine — it is a historical accident that has been conserved because changing it would cause catastrophic errors in all proteins.

The universality of the code is powerful evidence that all life on Earth shares a COMMON ANCESTOR — from whom this code was inherited.

Minor exceptions exist in some mitochondrial genomes and a few unicellular organisms — but these exceptions prove the rule: the standard code is overwhelmingly universal.

EXAM TIP

Drawing a nucleotide: always include all THREE components (phosphate + sugar + base). Missing one = lost mark.

Antiparallel means the strands run in OPPOSITE directions — one 5'→3', the other 3'→5'. This is not the same as saying they are 'upside down'. Explain it in terms of 5' and 3' ends.

Base pairing: A–T (2 H-bonds) and C–G (3 H-bonds). Know which has more — C–G is MORE stable (3 bonds). High GC content = higher melting temperature of DNA.

A2.2 — Cell Structure

Prokaryote vs Eukaryote · Organelles · Microscopy · Magnification · Atypical cells

The cell is the basic unit of life. All living organisms are made of cells, but cells come in two fundamental types: prokaryotes (simpler, smaller) and eukaryotes (more complex, with a membrane-bound nucleus and organelles). Understanding what each part does is essential for linking structure to function throughout the course.

Feature	Prokaryote	Eukaryote
Size	Typically 0.2–2 µm	Typically 10–100 µm (much larger)
Nucleus	No membrane-bound nucleus — DNA in NUCLEOID region	Membrane-bound nucleus with nuclear envelope and nuclear pores
DNA	Circular DNA in nucleoid; may have plasmids (small circular DNA)	Linear DNA associated with histone proteins; enclosed in nucleus
Ribosomes	70S (smaller — 50S + 30S subunits)	80S (larger — 60S + 40S subunits). Mitochondria/chloroplasts have 70S.
Membrane-bound organelles	NONE — no mitochondria, ER, Golgi, etc.	YES — mitochondria, ER, Golgi, lysosomes, vacuoles, etc.
Cell wall	Present — made of PEPTIDOGLYCAN (bacteria)	Present in plants (cellulose), fungi (chitin), absent in animal cells
Examples	Bacteria, Archaea	Animals, plants, fungi, protists
Cell division	Binary fission — simple splitting	Mitosis (and meiosis for sexual reproduction)

Organelle	Key structural features	Function — what it does	Memory link
Nucleus	Double membrane (nuclear envelope), nuclear pores, nucleolus inside	Stores DNA; site of transcription (mRNA synthesis); nucleolus makes rRNA	The 'headquarters' — instructions here
Rough ER	Membrane network continuous with nucleus; covered in ribosomes	Synthesis and initial modification of proteins destined for secretion	'Rough' because of ribosomes — protein factory
Smooth ER	Membrane network; no ribosomes	Lipid synthesis, detoxification of drugs/alcohol, Ca^{2+} storage	'Smooth' — lipid lab
Golgi apparatus	Stack of flattened membrane sacs (cisternae); vesicles bud off	Receives proteins from ER, modifies, sorts, packages into vesicles for secretion or to lysosomes	'Post office' — sorts and sends packages

Mitochondrion	Double membrane; inner membrane folded into CRISTAE; MATRIX inside	Site of aerobic respiration — produces ATP (Krebs cycle in matrix; ETC on inner membrane)	Power plant — makes ATP
Chloroplast (plants only)	Double membrane; THYLAKOIDS (stacked into GRANA) in STROMA	Site of photosynthesis — light reactions in thylakoids; Calvin cycle in stroma	Solar panel — captures light energy
Lysosome	Single membrane; contains hydrolytic enzymes (acid pH inside)	Intracellular digestion — breaks down worn organelles, foreign material, cell debris	'Recycling bin'
Vacuole	Single membrane (tonoplast in plants); large in plant cells, small/absent in animal cells	Water storage and turgor pressure (plants); maintains cell shape	Water storage tank
Ribosome	Not membrane-bound; 2 subunits; site of translation	Protein synthesis — reads mRNA, assembles amino acids into polypeptides	The assembly line

Microscopy — Light vs Electron Microscopy

Feature	Light microscopy (LM)	Electron microscopy (EM)
Radiation used	Visible light (wavelength ~400–700nm)	Electrons (wavelength ~0.005nm — much shorter)
Resolution	~200nm (limited by light wavelength)	~0.1nm — 1000× better than LM — can see individual molecules
Magnification	Up to ~×1500 usefully	Up to ×500,000+ (TEM) or ×100,000 (SEM)
Specimen state	Living or dead; can be stained with coloured dyes	Must be DEAD — specimen in vacuum; stained with heavy metals (e.g. osmium, lead)
Image	Coloured (or stained) — 2D or can be 3D (confocal)	Black and white (TEM) or 3D surface image (SEM)
Types	Simple, compound, fluorescence, confocal	TEM (transmission — internal structure), SEM (scanning — surface structure)
What can be seen	Cells, large organelles (nucleus, chloroplasts, mitochondria)	Ribosomes, membranes, virus particles, molecular structures

Magnification Calculation — THIS APPEARS ON EVERY PAPER

Magnification = image size / actual size

Actual size = image size / magnification

ALWAYS use the same units throughout. If image is measured in mm and actual size should be in µm:

Convert: 1 mm = 1000 µm. So image size in mm × 1000 = image size in µm.

Example: A cell appears 30mm in an image. Magnification is ×500. Actual size = 30mm / 500 = 0.06mm = 60 µm.

Scale bar method: if a scale bar shows '10 µm = 15mm on image', then magnification = 15mm / 0.01mm = ×1500.

⚠ Show your working every time — partial credit is available.

KEY DEFINITIONS

Atypical cells — cells that do not fit the standard eukaryote model:

Red blood cells (erythrocytes): no nucleus (expelled during maturation), no mitochondria — maximum space for haemoglobin, biconcave disc shape increases SA for gas exchange.

Platelets (thrombocytes): not true cells — cell FRAGMENTS from megakaryocytes. No nucleus. Role in blood clotting.

Viruses: NOT cells. No nucleus, no organelles, no cytoplasm, no cell membrane. Consist of nucleic acid (DNA or RNA) surrounded by a protein coat (capsid). Cannot reproduce independently — require a host cell. Not considered living by most definitions.

7 functions of life (MRS GREN): Metabolism, Response, Homeostasis, Growth, Reproduction, Excretion, Nutrition.

EXAM TIP

Prokaryote vs eukaryote — the most examined comparison. Know at least 5 differences. Key ones: no membrane-bound nucleus (prokaryote), 70S vs 80S ribosomes, no membrane-bound organelles (prokaryote), peptidoglycan cell wall (bacteria).

Organelle function questions: always link STRUCTURE to FUNCTION. Cristae increase the surface area of the inner mitochondrial membrane — more space for electron transport chain proteins — more ATP produced.

Magnification calculation: write the formula first, then substitute. Many students lose marks by doing the calculation correctly but not showing the formula.

A3.1 — Diversity of Organisms

Variation · Species · Taxonomy · Speciation · Karyotyping · Genomes

Life on Earth encompasses extraordinary diversity — estimated 8–10 million species. Yet this diversity is organised into patterns that reflect shared ancestry. This section covers how we classify and define organisms, and how new species arise.

Concept	Definition	Example / detail
Continuous variation	Variation that shows a range of values between two extremes — normal distribution (bell curve)	Height, mass, foot size — many genes (polygenic) + environment interact to produce a spectrum of values
Discontinuous variation	Variation that falls into distinct categories with no intermediate forms	ABO blood type (A, B, AB, O) — controlled by single gene with multiple alleles. Presence/absence of tongue-rolling.
Biological species concept	A species is a group of organisms that CAN interbreed and produce FERTILE offspring, and are REPRODUCTIVELY ISOLATED from other species	A horse and donkey can breed (mule) but the mule is infertile → horses and donkeys are different species
Binomial nomenclature	Two-part naming system: Genus species. Genus capitalised, species lowercase. Both italicised (underlined in handwriting).	Homo sapiens (modern human). Panthera leo (lion). Escherichia coli (common gut bacterium)
Allopatric speciation	Geographic isolation separates populations → gene flow stops → populations evolve independently → reproductive isolation develops → new species	Darwin's finches: populations isolated on different Galapagos islands → beak shape evolved differently → now distinct species
Sympatric speciation	New species arise within the same geographic area (without geographic barrier) — e.g. via polyploidy in plants or different mate preferences	Polyploidy in plants: chromosome doubling produces organisms that can no longer breed with the original species
Karyotype	The complete set of chromosomes of an organism, displayed in homologous pairs, arranged by size	Used to identify species, detect chromosomal abnormalities (e.g. Down syndrome = trisomy 21 — 47 chromosomes), sex determination

Taxonomy — Hierarchical Classification

Remember: Dead King Philip Came Over For Good Soup

Domain → Kingdom → Phylum → Class → Order → Family → Genus → Species

Each level is more specific. All members of the same genus share more features than members of the same family.

Three domains: Bacteria (prokaryotes, peptidoglycan walls), Archaea (prokaryotes, extremophiles, no peptidoglycan), Eukarya (all eukaryotes — plants, animals, fungi, protists).

Modern taxonomy is increasingly based on molecular evidence (DNA sequences) rather than morphology alone — phylogenetic classification reflects evolutionary relationships.

KEY DEFINITIONS

Genome: the complete set of genetic material (DNA) in an organism. Includes ALL genes plus non-coding sequences.

Genome size variation: C-value paradox — genome size does NOT directly correlate with organism complexity. Some plants and amphibians have larger genomes than humans. Reasons: different amounts of introns (non-coding DNA), repetitive sequences, transposons ('jumping genes'), gene duplication events.

Reproductive isolation: any mechanism that prevents gene flow between populations. Can be: geographic (physical barrier), temporal (breed at different times), behavioural (different courtship), mechanical (incompatible genitalia), gametic (gametes cannot fuse), hybrid infertility (offspring infertile).

EXAM TIP

Species definition exam question: ALWAYS include all three parts — (1) can interbreed, (2) fertile offspring, (3) reproductively isolated from other species. Missing any part loses marks.

Karyotype: humans have 46 chromosomes = 23 PAIRS. Gametes have 23 (haploid). Karyotyping detects: trisomy 21 (Down syndrome), Turner syndrome (45,X), Klinefelter syndrome (47,XXY).

Binomial nomenclature: Homo sapiens — notice capitalisation and italics. 'Homo Sapiens' = wrong. 'homo sapiens' = wrong. 'H. sapiens' is acceptable abbreviation after first use.

A4.1 — Evolution & Speciation

Lamarck vs Darwin · Natural selection · Evidence · Homologous/Analogous · Convergent evolution

Evolution is the change in heritable characteristics of a population over successive generations. Natural selection — Darwin's mechanism — is the central process driving evolution. Understanding evolution connects genetics, ecology, and biodiversity.

Theory	Proposed by	Core claim	Why it is wrong/incomplete
Use and Disuse / Inheritance of Acquired Characteristics	Jean-Baptiste Lamarck (early 1800s)	Organisms develop traits through use/disuse during their lifetime, and these acquired characteristics are passed to offspring. Classic example: giraffes stretched necks during lifetime, passed longer necks to offspring.	Acquired characteristics cannot be inherited — genetic changes only occur in germline cells. A bodybuilder's muscles are not inherited by their children. Epigenetic inheritance is a modern nuance but does not support Lamarckism.
Natural Selection	Charles Darwin (and Alfred Russel Wallace), 1858	Heritable variation exists within populations. Some variants are better suited to the environment. These individuals survive and reproduce more (differential reproductive success). Traits are inherited. Over time, favourable traits become more common — populations change.	The MECHANISM is correct and supported by overwhelming evidence. Darwin did not know about genetics (Mendel's work was unknown to him) — the 'Modern Synthesis' integrated natural selection with genetics.

PROCESS: Natural Selection — The Four Requirements

1. VARIATION: There must be heritable variation in the population — individuals differ in traits that can be inherited (e.g. coat colour, beak length).

2. OVERPRODUCTION: More offspring are produced than can survive — competition for limited resources (food, space, mates).

3. DIFFERENTIAL SURVIVAL (Survival of the Fittest): Individuals with traits that better suit the environment are more likely to SURVIVE and REPRODUCE — not necessarily the strongest, but the best-fitted to the specific environment.

4. INHERITANCE: Advantageous traits are passed from parent to offspring via genes — so the next generation has a higher frequency of the favourable trait.

5. RESULT: Over many generations, the frequency of advantageous alleles increases in the population — the population evolves.

Evidence type	Description	Example
Fossil record	Fossils show progressive change over time. Transitional fossils show intermediate forms between major groups.	Archaeopteryx — transitional between dinosaurs and birds (feathers + teeth). Horse evolution series showing gradual increase in size and toe reduction.
Comparative anatomy — Homologous structures	Structures with the SAME evolutionary origin but DIFFERENT function in different species — evidence of common ancestry (divergent evolution).	Pentadactyl limb: human hand, bat wing, whale flipper, horse leg — same bones (humerus, radius, ulna, carpals, phalanges) rearranged for different functions.
Comparative anatomy — Analogous structures	Structures with the SAME function but DIFFERENT evolutionary origin — evidence of convergent evolution (NOT of common ancestry).	Wings of birds vs wings of insects — both for flight but evolved independently from different ancestral structures.

Molecular evidence	DNA sequences, amino acid sequences of proteins — more similar sequences indicate more recent common ancestor.	Cytochrome c (protein in ETC): humans and chimpanzees have identical cytochrome c. Humans and yeast differ by ~50 amino acids — reflecting greater evolutionary distance.
Biogeography	Geographic distribution of species — patterns explained by evolution and continental drift.	Marsupials concentrated in Australia (isolated continent) — evolved independently. Darwin's finches on Galapagos show adaptive radiation.

KEY DEFINITIONS

Convergent evolution: when unrelated species independently evolve SIMILAR features in response to similar environmental pressures. The features are ANALOGOUS (same function, different origin). Example: dolphins (mammals) and sharks (fish) both evolved streamlined bodies for swimming — convergent evolution.

Adaptive radiation: rapid evolution of many new species from a single ancestor — each adapting to a different ecological niche. Example: Darwin's finches on Galapagos islands, each with a beak adapted to different food sources.

Types of natural selection: Stabilising (extremes selected against — most common in stable environments), Directional (one extreme favoured — population shifts in one direction), Disruptive (both extremes favoured — intermediate eliminated — can lead to speciation).

EXAM TIP

Homologous vs analogous: HOMOLOGOUS = same structure, different function (evidence of COMMON ANCESTRY — divergent evolution). ANALOGOUS = same function, different structure (evidence of CONVERGENT EVOLUTION — NOT common ancestry). This distinction is frequently examined.

Natural selection exam question: describe the MECHANISM in 4 steps. Markers look for: variation, differential survival, heritability, change over time. Include specific examples where possible.

Don't confuse 'evolution' with 'natural selection'. Natural selection is the MECHANISM by which evolution occurs. Evolution is the result (change in allele frequencies over time).

A4.2 — Conservation of Biodiversity

Three levels · Anthropogenic threats · EDGE · Ex situ vs in situ

Biodiversity refers to the variety of life on Earth. It is being lost at an unprecedented rate due to human activity. Understanding what biodiversity is, why it matters, and how we can protect it is a growing area of IB Biology.

Level of biodiversity	What it encompasses	Why it matters	Example
Genetic diversity	Variety of alleles within a species — the total genetic variation in a population's gene pool	Larger gene pool = greater ability to adapt to environmental change. Low genetic diversity = vulnerable to disease and environmental change (e.g. cheetahs — severe genetic bottleneck)	Cheetahs: all genetically nearly identical → highly vulnerable. Wild crop varieties maintain genetic diversity for future breeding programmes.
Species diversity	Variety of different species in an ecosystem — includes both species richness (number) and evenness (relative abundance)	High species diversity = more stable ecosystems. More complex food webs. Greater ecosystem resilience. Many species provide services to humans.	Tropical rainforest: very high species diversity. Arctic tundra: low species diversity. Loss of one keystone species can destabilise the whole ecosystem.
Ecosystem diversity	Variety of different ecosystems or habitats within a region or on Earth	Different ecosystems provide different services. Coastal wetlands, rainforests, coral reefs all unique.	Tropical rainforests, coral reefs, deep ocean, tundra — each unique community of species and physical environment.

Anthropogenic threat	Mechanism	Example
Habitat loss and fragmentation	Deforestation, agriculture, urbanisation destroy ecosystems. Fragmentation isolates populations — reduces gene flow, increases vulnerability.	Amazon deforestation for cattle ranching and soya agriculture — destroys habitat for jaguars, tapirs, thousands of insect species.
Invasive species	Non-native species introduced (deliberately or accidentally) — outcompete native species, alter ecosystems.	Cane toads in Australia (introduced to control pests) — toxic, killed native predators. Brown tree snake in Guam — eliminated 10 of 12 native bird species.
Overexploitation	Hunting, fishing, collecting beyond sustainable rates — populations cannot recover.	Collapse of Atlantic cod fisheries. Commercial whaling brought many whale species to near-extinction.
Pollution	Pesticides, heavy metals, plastic, light pollution, noise pollution harm species directly or disrupt behaviour.	DDT caused eggshell thinning in raptors (biomagnification). Plastic ingestion kills seabirds and marine turtles.
Climate change	Rising temperatures, changing precipitation, sea level rise — shifts habitats, disrupts phenology, bleaches coral, melts ice.	Polar bears: loss of sea ice reduces hunting platforms. Coral bleaching: mass mortality events on Great Barrier Reef.

KEY DEFINITIONS

EDGE programme (Evolutionarily Distinct and Globally Endangered): prioritises conservation of species that are both evolutionarily unique (few close relatives — 'distinct' branch of the tree of life) AND globally endangered. These species represent irreplaceable evolutionary history. Examples: Chinese giant salamander, Aye-aye (Madagascar lemur).

In situ conservation: protecting species IN THEIR NATURAL HABITAT — national parks, reserves, protected areas, wildlife corridors. Maintains ecological interactions. More sustainable long-term but habitat must exist and be protected.

Ex situ conservation: protecting species OUTSIDE their natural habitat — zoos, botanical gardens, seed banks, captive breeding programmes. Useful when habitat is destroyed. Maintains genetic diversity. Problems: animals may lose survival skills, small populations in captivity.

Ecosystem services: benefits humans obtain from ecosystems — provisioning (food, water, medicine), regulating (climate, flood, pollination), cultural (tourism, spiritual, aesthetic), supporting (nutrient cycling, soil formation).

Intrinsic value: species have value in themselves, regardless of human benefit. Extrinsic/instrumental value: value to humans (economic, ecological, aesthetic).

EXAM TIP

For conservation questions: state the THREE LEVELS of biodiversity (genetic, species, ecosystem). Many students only mention species diversity.

EDGE programme: know both criteria — EVOLUTIONARILY DISTINCT (unique, few close relatives) AND GLOBALLY ENDANGERED. Both must be present for EDGE priority.

In situ vs ex situ: know at least 2 advantages AND 1 disadvantage of each. Best answers discuss how the two approaches are COMPLEMENTARY.

LINKS TO OTHER TOPICS

A1.1 Water → B2.1 (solvent properties of water for transport), C1.2 (water produced in aerobic respiration), D2.3 (water potential and osmosis)

A1.2 Nucleic Acids → D1.1 (DNA replication uses nucleotides), D1.2 (transcription/translation), D1.3 (mutations)

A2.2 Cell Structure → B2.2 (organelles in detail), C1.2 (mitochondria in respiration), C1.3 (chloroplasts in photosynthesis), D2.1 (mitosis/meiosis)

A3.1 Diversity → A4.1 (speciation produces diversity), A4.2 (conservation of biodiversity)

A4.1 Evolution → D4.1 (natural selection detail), D3.2 (inheritance underpins genetic variation)

A4.2 Conservation → D4.2 (ecosystem stability), D4.3 (climate change threats to biodiversity)

THEME A — EXAM READINESS CHECKLIST

A1.1: **WATER**

- ☑ Name 4+ properties of water and explain each in terms of H-bonding
- ☑ Give a biological consequence for each property
- ☑ Explain why ice floats

A1.2: **NUCLEIC ACIDS**

- ☑ Draw/describe nucleotide structure
- ☑ Compare DNA vs RNA
- ☑ State complementary base pairs and number of H-bonds
- ☑ Explain antiparallel
- ☑ State significance of universal genetic code

A2.2: **CELL STRUCTURE**

- ☑ List 5+ differences between prokaryotes and eukaryotes
- ☑ State function of each organelle
- ☑ Compare LM vs EM
- ☑ Calculate magnification
- ☑ Describe atypical cells

A3.1: **DIVERSITY OF ORGANISMS**

- ☑ Biological species concept (all 3 parts)
- ☑ Binomial nomenclature rules
- ☑ Taxonomy levels in order
- ☑ Allopatric vs sympatric speciation
- ☑ What karyotypes show

A4.1: **EVOLUTION & SPECIATION**

- ☑ Lamarck vs Darwin — why Lamarck was wrong
- ☑ 4 steps of natural selection
- ☑ Homologous vs analogous structures
- ☑ Types of evidence for evolution

A4.2: **CONSERVATION OF BIODIVERSITY**

- ☑ 3 levels of biodiversity
- ☑ 5 anthropogenic threats
- ☑ EDGE programme criteria
- ☑ In situ vs ex situ conservation
- ☑ Ecosystem services